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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/188,967	03/01/2021	Justin Wade Hart	148571.04557	6436

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QUARLES & BRADY LLP/Emerson Automation
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EXAMINER	
COLON MORALES, DAVID	

ART UNIT	PAPER NUMBER
3753	

NOTIFICATION DATE	DELIVERY MODE
10/08/2021	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

pat-dept@quarles.com

Office Action Summary	Application No. 17/188,967	Applicant(s) Hart et al.	
	Examiner DAVID COLON-MORALES	Art Unit 3753	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 03/01/2021.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-7,9-16 and 18-20 is/are rejected.

8) ☑ Claim(s) 8 and 17 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 03/01/2021 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☑ Other: PTO-2333 Notice of Considerations.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Drawings

2. The drawings are objected to as failing to comply with 37 CFR 1.84(p)(5) because they do not include the following reference sign(s) mentioned in the description:

a. "Stem assembly **338**" mentioned in at least Para. [0055] is not shown in the drawings. The Office suggest that reference number 338 is added to at least Fig. 3a.

3. Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Specification

4. The lengthy specification has not been checked to the extent necessary to determine the presence of all possible minor errors. Applicant's cooperation is requested in correcting any errors of which applicant may become aware in the specification.

Claim Interpretation

5. The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

6. The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term "means" or "step" or a term used as a substitute for "means" that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word “means” (or “step”) are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word “means” (or “step”) are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

Claim Rejections - 35 USC § 102

7. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

8. Claim(s) 1, 5-7 and 9 is/are rejected under 35 U.S.C. 102a1 as being anticipated by Rheinauer (DE 29506395).

9. **Regarding claim 1, Rheinauer (DE 29506395) teaches in at least Figs. 1-3** of a pressure regulator (pressure regulator 1) comprising: a valve body (housing 4) that defines a fluid flow path (see at least Fig. 1) between an inlet (inlet 2) and an outlet (outlet 3); an orifice assembly (annular valve seat 11 comprising an orifice that is controlled by valve body 13) that is positioned along the fluid flow path; a stem (see the stem assembly comprising at least control rod 18 and cylinder part 15); a lever (lever

26) configured to control movement of the stem when engaged with the stem, as driven by movement of a diaphragm (diaphragm 8); a control member (valve body 13 acting as a valve to control the flow) that is coupled to the stem, wherein, in a first mode of operation of the pressure regulator (normal/regular operation), the control member is moveable relative to the orifice assembly between a first position (the fully closed position, Fig. 3) in which the control member contacts the orifice assembly to restrict fluid flow along the fluid flow path and a second position (the fully opened position, not shown but understood to be a position where the valve body is lifted away from the seat but has yet to reach the position of Fig. 2 thus allowing full flow) in which the control member is separated from the orifice assembly; and a mechanical stop (first O-ring 16 act as a mechanical stop and valve for the valve assembly since it engages with sealing surface 11a preventing further outward movement of the valve 10 and preventing a full flow situation) that is coupled to the stem, wherein the mechanical stop is configured to engage a stop feature (sealing surface 11a) in a second mode of operation of the pressure regulator (such as when a main gasline is shutoff, see at least Para. [0053]) to prevent the control member from moving past a third position (see Fig. 2, notice that the first O-ring 16 act as a mechanical stop when it engages with the sealing surface 11a which prevents further outward movement of the valve 10 past the third position of Fig. 2, see at least Para. [0035-0056] for more details of the operation of the device of Rheinauer). **Thus, the device of Rheinauer meets all the limitations of claim 1.**

10. **Regarding claim 5** and the limitation of the pressure regulator of claim 1, wherein the mechanical stop is formed as a ring that at least partially surrounds the stem; the device of Rheinauer meets this limitation as shown in at least Fig. 2 with the

O-ring 16 being an ring/elongate-sleeve having a length/dimension for sealingly engaging with sealing surface 11a during the second mode.

11. **Regarding claim 6** and the limitation of the pressure regulator of claim 5, wherein the mechanical stop is formed as an elongate sleeve; the device of Rheinauer meets this limitation as shown in at least Fig. 2 with the O-ring 16 being an ring/elongate-sleeve having a length/dimension for sealingly engaging with sealing surface 11a during the second mode.

12. **Regarding claim 7** and the limitation of the pressure regulator of claim 6, wherein the elongate sleeve is configured to be secured to the stem at a fixed location on the stem, so that a length of the elongate sleeve determines a distance between the second and third positions of the control member; the device of Rheinauer meets this limitation as shown in at least Fig. 2 with the O-ring 16 being installed on groove 15a of the stem assembly and having a length/dimension for sealingly engaging with sealing surface 11a during the second mode which engagement determines the distance between the second and third position of the valve body 13.

13. **Regarding claim 9** and the limitation of the pressure regulator of claim 1, wherein the stem extends through a wall of the valve body between the orifice assembly and the lever; and wherein the mechanical stop is disposed on the stem opposite the wall from the lever; the device of Rheinauer meets this limitation as shown in at least Fig. 1-3 with stop 16 is disposed on the control valve 10 side as opposed to the lever/diaphragm 5 side.

Claim Rejections - 35 USC § 103

14. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

15. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

16. The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

17. This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly

owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

18. Claim 11 is/are rejected under 35 U.S.C. 103 as being unpatentable over Rheinauer (DE 29506395).

19. **Regarding claim 11, the device of Rheinauer teaches** the limitation of the pressure regulator of claim 1, wherein the mechanical stop is formed separately from the stem (notice in at least Fig. 2 that the O-ring 16 is a separate component from the stem assembly). **Rheinauer fails to explicitly disclose** the limitations of "wherein the mechanical stop is configured to be installed onto the stem **after installation of the pressure regulator for operation**". However, notice that this limitation is a Product-by-Process Limitation (in this case, the product is the mechanical stop or O-ring 16 and the process is the step of its installation after installation of the pressure regulator). The patentability of a product does not depend on its method of production. If the product in the product-by-process claim is the same as or obvious from a product in the prior art, the claim is unpatentable even though the prior product was made by a different process (see MPEP 2113). Additionally, notice in at least Figs. 1-2 that the O-ring 16 is configured to be capable of being either pre-installed with the stem assembly or be later installed and/or serviced once the stem assembly is installed via the access provided at the safety shut-off valve 6 side. As such, it would have been obvious to one of ordinary skill in the art at the time that the invention was filed to install the O-ring 16 to the stem assembly either before the pressure regulator is ready for operation or after is ready for

operation since the device is capable of being serviced via the access provided at the safety shut-off valve 6 side allowing a user to service the component should it wear off.

Double Patenting

20. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for

applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

21. **Claims 1-7, 9, 11-16 and 18-20 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-20 of U.S. Patent No. 11,022,988.**

22. Although the claims at issue are not identical, they are not patentably distinct from each other because the device of the patent either explicitly or implicitly anticipates all the limitations of the present application and/or any missing structure or detail would be obvious over the cited prior art. Notice that the claims of the current application recites limitations directed to a pressure regulator comprising at least: a valve body, an orifice assembly, a stem, a lever, a diaphragm, a control member, a mechanical stop and a stop feature. Similarly, claims 1-20 of the patent recites limitations directed to a pressure regulator comprising at least: a valve body, an orifice assembly, a stem, a lever, a diaphragm, a primary control member and a secondary control member. The difference between the claims of the present application and the claims of the parent is

the recitations of the mechanical stop, the stop feature and the claim scope being different. However, notice that at least claim 8 of the parent case discloses the secondary control member and a second side of the orifice assembly acting as a “stop” and “stop feature”, respectively. Following the rationale in In re Goodman cited in the preceding paragraph, where the applicant has once been granted a patent containing a claim for the specific or narrower invention, the applicant may not then obtain a second patent with a claim for the generic or broader invention without first submitting an appropriate terminal disclaimer. Note that since the claims of the application are anticipated by the claims of the patent and since anticipation is the epitome of obviousness, then the claims of the application are obvious over the claims of the patent. Additionally, any missing structure or detail would be obvious over the cited prior art with at least Rheinauer (DE 29506395) teaching of a similar pressure regulator comprising similar structure and function as the claimed inventions (see the rejection above for details), in particular, comprising an O-ring 16 capable of acting as a mechanical stop when it engages a stop feature (sealing surface 11a) of the orifice assembly (seat 11) which allows the valve to have three positions (a first position/fully closed position, a second position/fully opened position and a third position/safety posiiton) with the third position aiding in preventing full flow from occurring during an irregular event such as main gas line shutoff event.

Allowable Subject Matter

23. **Claims 12-16 and 18-20** are allowed pending resolution of the double patenting rejection addressed above.

24. **Claims 2-4 and 10** are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims and pending resolution of the double patenting rejection addressed above.

25. **Claim 8 and 17** is objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

26. The following is a statement of reasons for the indication of allowable subject matter:

The closest prior art are Rheinauer (DE 29506395), Sydney (US 3,580,271), Pommersheim (US 3,042,064), Pierre (CH 574068), Johnson (US 3,892,255), Hughes (US 3,488,685) and Hawkins (US 6,968,857).

Pommersheim, Hughes and Hawkins teaches the general state of the art with pressure regulators having similar structure and function as applicant's general invention. **Johnson, Pierre, Sydney and Rheinauer teaches** of other examples of pressure regulators that comprises a safety feature to prevent unintentional full open position of the valve similar to a key feature of the claimed invention. In particular, Rheinauer teaches of an O-ring capable of servicing as a mechanical stop similar to a key feature of the claimed invention. Notice that the prior art fails to disclose in the second mode of operation wherein the lever is disconnected from the stem and/or the stem moves freely relative to the lever (notice that applicant's invention is directed to a pressure regulator assembly capable of limited functionality even if the connection between the lever and the stem is

broken which is different to the prior art which provides a safety means to prevent the full open position during different scenarios) and the prior art fails to disclose the particular structure, location and function of mechanical stop and the stop feature as claimed. **As such, the closest prior art fails to disclose or render obvious without impermissible hindsight** the particular structure and function of the pressure regulator (100) comprising the valve body (104), the orifice assembly (126), the stem (336), the lever (338), the diaphragm (144), the control member (332), the first mode of operation of the pressure regulator (normal operation, wherein the lever is connected to the stem and/or the stem is moved by the lever), the second mode of operation (when the lever is disconnected from the stem and/or the stem moves freely relative to the lever), the mechanical stop (334) and the stop feature (118) in combination with all the limitations as claimed in claims 2-4, 8, 10, 12-20 and as shown in at least Figs. 3a-3d of the application.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to David Colon-Morales, whose telephone number 571-270-1741 and fax number is 571-270-2741. The examiner can normally be reached on Monday-Friday (7:30AM-3:30PM EST).

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisors can be reached by phone. **Craig Schneider** can be reached at 571-272-3607 or **Mary McManmon** can be reached at 571-272-6007 or **Kenneth Rinehart** can

be reached at 571-272-4881. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/DAVID COLON-MORALES/
Primary Examiner, Art Unit 3753